Chapter 6, when gravity matters— as it did in the dense early universe—homogeneity implies low entropy. That's a good thing, because our discussion of time's arrow relied heavily on the universe's starting out with low entropy. One of our goals in this part of the book is to go as far as we can toward explaining this observation—we want to understand how the homogeneous, low-entropy, highly unlikely environment of the early universe came to be. This would take us a big step closer to grasping the origin of time's arrow.

Second, although the universe has been evolving since the big bang, on average the evolution must have been nearly identical across the cosmos. For the temperature here and in the Whirlpool galaxy, and in the Coma cluster, and everywhere else to agree to four decimal places, the physical conditions in every region of space must have evolved in essentially the same way since the big bang. This is an important deduction, but you must interpret it properly. A glance at the night sky certainly reveals a varied cosmos: planets and stars of various sorts sprinkled here and there throughout space. The point, though, is that when we analyze the evolution of the entire universe we take a macro perspective that averages over these "small"-scale variations, and large-scale averages *do* appear to be almost completely uniform. Think of a glass of water. On the scale of molecules, the water is extremely heterogeneous: there is an H_2O molecule over here, an expanse of empty space, another H_2O molecule over there, and so on. But if we average over the small-scale molecular lumpiness and examine the water on the "large" everyday scales we can see with the naked eye, the water in the glass looks perfectly uniform. The nonuniformity we see when gazing skyward is like the microscopic view from a single H_2O molecule. But as with the glass of water, when the universe is examined on large enough scales—scales on the order of hundreds of millions of light-years—it appears extraordinarily homogeneous. The uniformity of the radiation is thus a fossilized testament to the uniformity of both the laws of physics and the details of the environment across the cosmos.

This conclusion is of great consequence because the universe's uniformity is what allows us to define a concept of time applicable to the universe as a whole. If we take the measure of change to be a working definition of elapsed time, the uniformity of conditions throughout space